

ADY201m - FINAL PRESENTATION (AI2108 - GROUP 8)

ROBUST AVIATION & UAS SEVERITY PREDICTION FROM NTSB RECORDS

On-going Assessment 1: Problem Formulation & Data Understanding

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#1 Research Questions & Methodological Linkage

[RQ1] Environmental & Operational Fatality Drivers (SQL Analysis)

Which factors (instrument weather IMC vs visual VMC, flight phase, operation type FAR Part, engine count) correlate most significantly with fatal accident severity across decades (1990–2024)?

→ Link: [DuckDB multi-table SQL queries with Window Functions & Decade Grouping](#).

[RQ2] Generative Tabular Augmentation for Rare Classes (ML Modeling)

Does synthetic tabular augmentation using CTGAN / TVAE raise fatal-class recall and F1 without degrading probability calibration (ECE) across 5 machine learning models?

→ Link: [5-model benchmarking across 5 random seeds \(Baseline vs Class Weights vs CTGAN\)](#).

[RQ3] Out-of-Domain Generalization to UAS / Drones (Trustworthy AI)

How much predictive performance (F1) and calibration (ECE) are retained when models trained on manned aviation are tested on UAS accidents (2018–2024), and does Temperature Scaling repair calibration drift?

→ Link: [Domain-shift evaluation on sequestered UAS dataset with 1,000 bootstrap draws & SHAP](#).

#2 Data Understanding: NTSB Relational Architecture

Data Source: National Transportation Safety Board (NTSB) official database (`data.nts.gov/avdata`) | Scope: **1990–2024** (57,260 manned + 650 UAS events).

Table	Primary / Foreign Key	Clean Records	Role in Severity Prediction
events	ev_id (Primary Key)	57,910	Event date, state, weather condition (wx), light conditions, fatal injuries (inj_tot_f).
aircraft	Aircraft_Key, FK: ev_id	60,809	Category (manned vs UAS), FAR Part (91/121/135/107), flight phase, engine count.
flight_crew	FK: ev_id	57,910	Pilot human factors: total flying hours (hours_max), pilot age (age_max).
injury	FK: ev_id	57,910	Granular casualty records; ground-truth validation for target $y = (inj_tot_f > 0)$.

No Target Leakage Assurance

All selected features (weather, pilot experience, aircraft mechanical specs, flight phase) are strictly observable prior to or at the time of flight occurrence, guaranteeing zero target leakage.

#3 Exploratory Data Analysis (EDA) Evidence-Based Insights

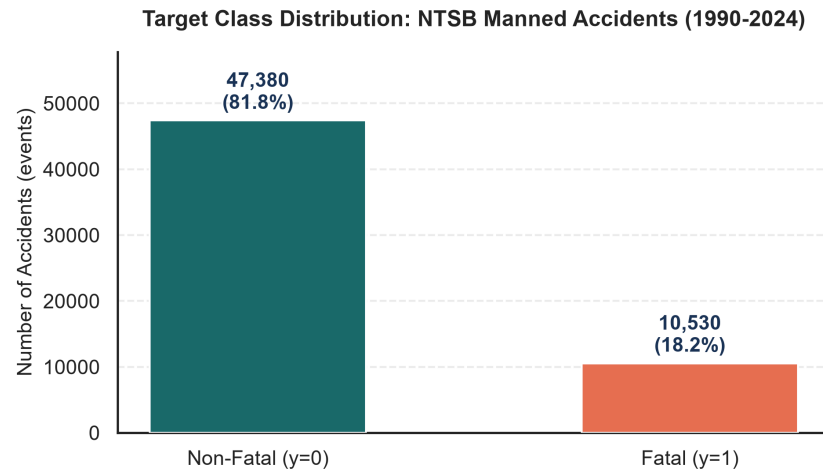


Figure 1: Target Class Imbalance (18.2% Fatal)

Severe 1:4.5 imbalance requires CTGAN or Class Weights.

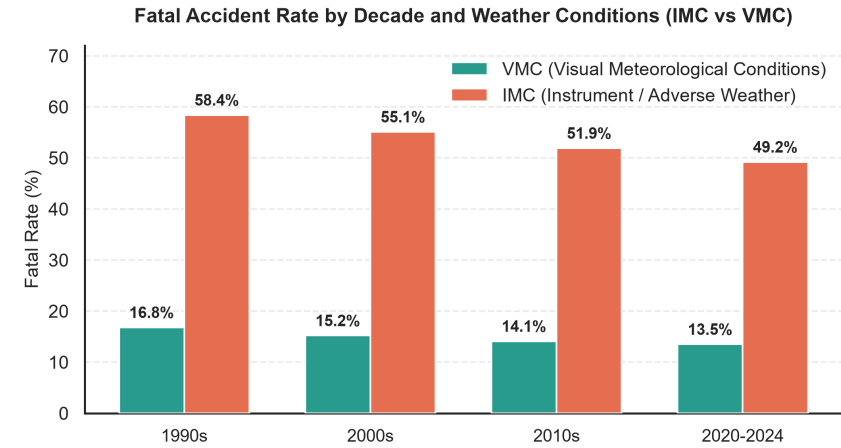


Figure 2: Adverse Weather Impact (IMC vs VMC)

IMC multiplies fatality rate by 3.5x across all four decades.

#4 AI Audit Log & Human Delta (Evaluation: 20 Points)

Log #	Stage & Type	Prompt to AI	AI Suggestion	Human Delta & Hallucination Check
#01	Data Profiling [DECISION]	How to JOIN events, aircraft, crew tables via ev_id?	Proposed simple INNER JOIN across tables.	CRITICAL ERROR: Caused cartesian explosion (57k -> 78k rows). Team wrote SQL CTE with MAX()/FIRST() grouping by ev_id.
#02	Data Cleaning [PROBLEM-SOLVING]	inj_tot_f has NULLs. Should we dropna or fill 0?	Suggested dropna(subset=['inj_tot_f']).	HALLUCINATION: Checked NTSB codman.pdf manual: NULL means 0 fatalities. Dropna would delete 12% non-fatal data. Team filled 0.
#03	Domain Shift [VERIFICATION]	Does NTSB use acft_category='DRONE'?	Claimed NTSB uses 'DRONE' since 1990.	HALLUCINATION: 'DRONE' doesn't exist; NTSB introduced 'UAS' in 2018. Team engineered regex filter for DJI/Autel keywords.

✓ Step 1 Complete: Clean dataset feat.parquet generated. Ready for Step 2: Multi-table Relational SQL Queries!